

Assignment 2

Issued: Sunday 27th September, 2026

Due: Sunday 18th October, 2026

Total: 100 points, plus 10 bonus points.

1. Matrix Diagonalizability

(25 points)

An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors.

- (a) For each matrix below, find the eigenvalues and a basis for each eigenspace by solving $(A - \lambda I)x = 0$. Then determine whether the matrix is diagonalizable and justify your answer.

Hint: The first matrix is block upper triangular. The determinant of a block triangular matrix is the product of the determinants of its diagonal blocks.

- i. (10 points)

$$A_1 = \begin{bmatrix} 2 & 3 & 0 \\ 1 & 4 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

- ii. (10 points)

$$A_2 = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

- (b) (5 points) Neither diagonalizability nor invertibility implies the other. Determine whether each of the two matrices above is invertible, and briefly explain how the two examples illustrate this statement.

2. Matrix Norms

(35 points)

This problem introduces matrix norms, a useful set of tools for measuring matrix size and approximation error in machine learning. We will also derive the Cauchy–Schwarz inequality for Euclidean vectors and establish useful bounds involving the spectral and Frobenius norms. These results provide a foundation for constructing similarity measures and proving error bounds for learning algorithms.

Let $x, y \in \mathbb{R}^n$, with $\langle x, y \rangle = x^T y$ and $\|x\|_2 = \sqrt{\langle x, x \rangle}$.

- (a) (10 points) For a matrix $A \in \mathbb{R}^{m \times n}$, define its *spectral norm* by

$$\|A\|_2 = \max_{z \neq 0} \frac{\|Az\|_2}{\|z\|_2}.$$

Show that $\|A\|_2 = \sigma_1$, the largest singular value of A .

- (b) (10 points) Similarly, define the *Frobenius norm* by

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2} = \sqrt{\text{tr}(A^\top A)}.$$

If $r = \text{rank}(A)$, show that

$$\|A\|_F^2 = \sum_{i=1}^r \sigma_i^2.$$

In other words, the squared Frobenius norm is the sum of the squared singular values of A .

- (c) (8 points) Prove the Cauchy–Schwarz inequality

$$|\langle x, y \rangle| \leq \|x\|_2 \|y\|_2.$$

- (d) (7 points) Prove that, for every $x \in \mathbb{R}^n$,

$$\|Ax\|_2 \leq \|A\|_2 \|x\|_2 \leq \|A\|_F \|x\|_2.$$

Clearly indicate where you use the definition of the spectral norm and where you use the relationship between the spectral and Frobenius norms.

3. Best Rank- k Approximation

(40 + 10 bonus points)

In class, we introduced the truncated SVD as a way to approximate a matrix using a lower-rank matrix. This problem explains why it is optimal: no other matrix of rank at most k can achieve a smaller error.

Let $A \in \mathbb{R}^{m \times n}$ have rank r and singular value decomposition $A = \sum_{i=1}^r \sigma_i u_i v_i^\top$, where $\sigma_1 \geq \dots \geq \sigma_r > 0$. For $k < r$, recall that the rank- k truncated SVD is defined by $\hat{A}_k = \sum_{i=1}^k \sigma_i u_i v_i^\top$.

- (a) (15 points) For the matrix

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix},$$

find a singular value decomposition and use it to compute the best rank-1 approximation of A .

- (b) (10 points) Show that the squared Frobenius-norm approximation error of $\hat{A}_k$ is the sum of the squares of the discarded singular values; that is,

$$\|A - \hat{A}_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2.$$

- (c) (8 points) Directly from the definition of the spectral norm, show that the spectral-norm approximation error of $\hat{A}_k$ is the $(k+1)$ -st largest singular value of A ; that is,

$$\|A - \hat{A}_k\|_2 = \sigma_{k+1}.$$

- (d) (10 points (bonus)) Let $B \in \mathbb{R}^{m \times n}$ satisfy $\text{rank}(B) \leq k$. Prove that

$$\|A - B\|_2 \geq \sigma_{k+1}.$$

Hint: Let $V_{k+1} = [v_1, \dots, v_{k+1}]$. Assume for contradiction that $\|A - B\|_2 < \|A - \hat{A}_k\|_2 = \sigma_{k+1}$. Use the rank-nullity theorem to show that there is a nonzero vector in $\text{null}(B) \cap \text{col}(V_{k+1})$, and then derive a contradiction.

- (e) (7 points) Using part (c) and the inequality stated in part (d), conclude that

$$\hat{A}_k \in \arg \min_{\text{rank}(B) \leq k} \|A - B\|_2.$$

You may use the inequality stated in part (d) even if you did not complete the bonus proof. Briefly explain what this result says about the truncated SVD.

Remark: The analogous optimality result also holds for the Frobenius norm. The proof is similar, so we do not ask you to write it here.